

## 1.7 Summarizing a distribution using moments and quantiles

To fully characterize the distribution of a random variable, we often write down objects like the cdf or the pmf/pdf, from which any probability can be computed. Having access to a complete description of the probability behavior of a random variable is nice, but we are often interested in extracting from the distribution of a random variable a few principle *features* or *summaries* of the distribution. We have already seen some:

- The expected value  $\mu = \mathbb{E}[X]$  is a single number that averages over the distribution of  $X$  to answer “for all the random variation in  $X$ , where, at the end of the day, will its value typically be concentrated?”
- The variance  $\sigma^2 = \text{var}(X) = \mathbb{E}[(X - \mu)^2]$  is a single number that answers “how far should I expect  $X$  to be from its expected value?”

When they exist, these are nice ways to concisely report on the key features of the random variation of  $X$ , without having to confront every detail of what the pmf/pdf is up to. In this lecture, we discuss a few more ways to summarize a distribution.

First, we can talk about the *shape* of the distribution; see Figure 1.27.

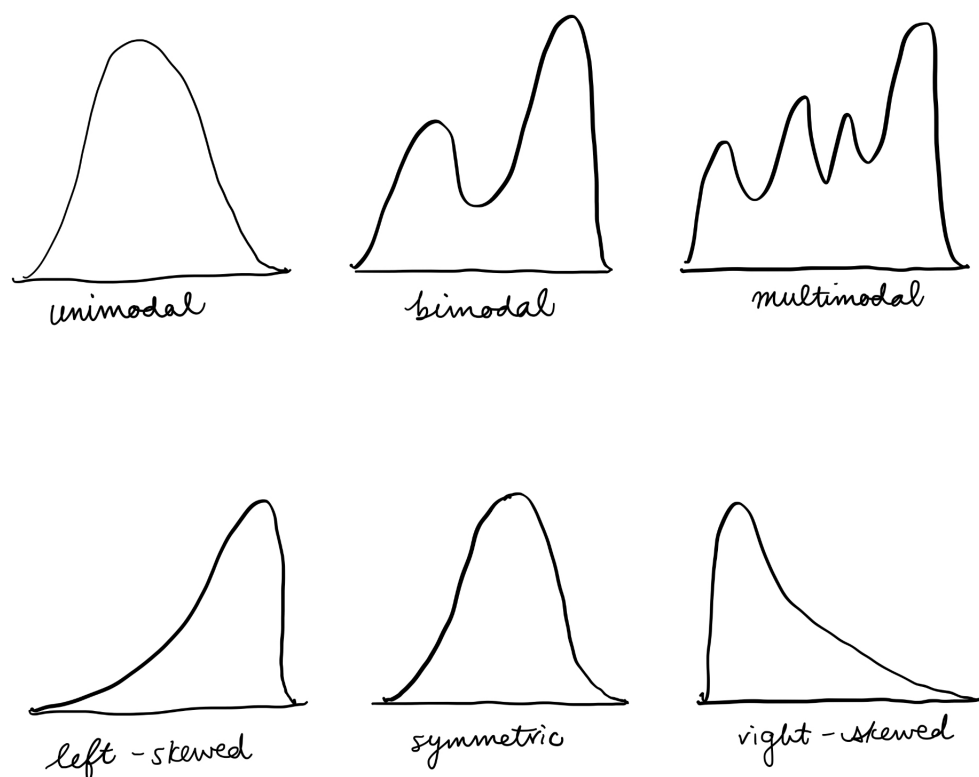


Figure 1.27: Shapes a distribution can take. Is there skew? Are there one or several modes? Note: left-skewed is also called “positively skewed” and right-skewed is also called “negatively skewed”.

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**Example 1.19.** • People’s heights form a bimodal distribution: women’s heights are the left bump (to the left on the  $x$ -axis) and men’s heights are the right bump (to the right on the  $x$ -axis).

- If someone doesn’t show up for an exam and gets a zero, this can make the exam grade distribution left-skewed.
- Incomes in the U.S. are a right-skewed distribution (with Elon Musk and co. in the upper tail).

As you can see from the figure, the skew characterizes the degree of symmetry of a distribution. The below definition gives a quantitative measure of skew. We also introduce another way to characterize a distribution called the kurtosis.

**Definition 1.20.** For a random variable  $X$  with mean  $\mathbb{E}[X] = \mu$  and variance  $\text{var}(X) = \sigma^2$ , the **skew** and **kurtosis** of  $X$  are defined as

$$\text{skew}(X) = \mathbb{E} \left[ \left( \frac{X - \mu}{\sigma} \right)^3 \right], \quad \text{kurtosis}(X) = \mathbb{E} \left[ \left( \frac{X - \mu}{\sigma} \right)^4 \right]. \quad (1.29)$$

Some remarks:

- While skew measures degree of asymmetry, kurtosis measure how heavy the tails of a distribution are. See Figure 1.28. Thicker tails mean extreme events are more likely.
- Skew and kurtosis are the third and fourth moments of the *standardized* version of  $X$ . We saw standardization in the last lecture in the context of Gaussian r.v.’s: if  $X \sim \mathcal{N}(\mu, \sigma^2)$ , then  $(X - \mu)/\sigma \sim \mathcal{N}(0, 1)$ . More generally, we can take any  $X$  with mean  $\mu$  and variance  $\sigma^2$ , and standardize it to have mean 0 and variance 1 by taking  $(X - \mu)/\sigma$ .
- The fact that we have standardized  $X$  before taking moments means that skew and kurtosis do not care about location or scale. Shifting the distribution or spreading the points out does not affect skew or kurtosis.

### 1.7.1 Moments

We have seen lots of variations on “moments” so far. Here are the definitions for the different moments of a random variable  $X$  that has  $\mathbb{E}[X] = \mu$  and  $\text{var}(X) = \sigma^2$ :

|                                               |                                                            |
|-----------------------------------------------|------------------------------------------------------------|
| $n^{\text{th}}$ raw moment                    | $E(X^n)$                                                   |
| $n^{\text{th}}$ absolute moment               | $E( X ^n)$                                                 |
| $n^{\text{th}}$ central moment                | $E[(X - \mu)^n]$                                           |
| $n^{\text{th}}$ absolute, central moment      | $E( X - \mu ^n)$                                           |
| $n^{\text{th}}$ standardized moment           | $E \left[ \left( \frac{X - \mu}{\sigma} \right)^n \right]$ |
| $n^{\text{th}}$ absolute, standardized moment | $E \left[ \left  \frac{X - \mu}{\sigma} \right ^n \right]$ |

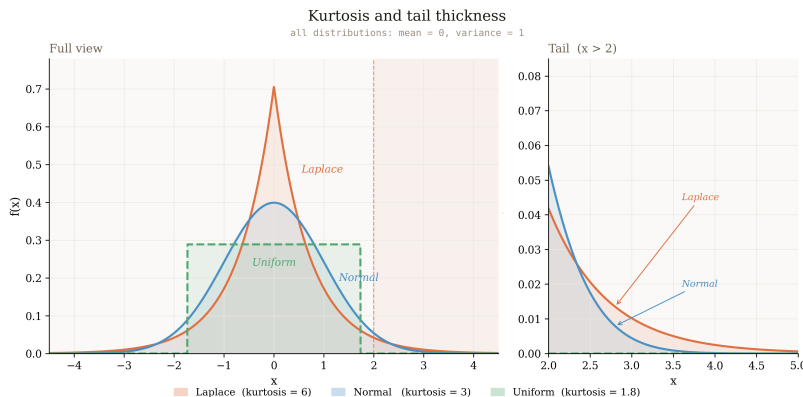


Figure 1.28: Higher kurtosis means heavier tails. The inset zooms in on the right tail, which is heavier for the Laplace distribution. The uniform distribution has the lightest tail, since the probability drops down to zero outside an interval.

So we see that the mean is the first raw moment, the variance is the second central moment, the skewness is the third centralized moment, and the kurtosis is the fourth standardized moment. This is summarized in Table 1.4.

|          | raw                             | central                                   | standardized                                                        |
|----------|---------------------------------|-------------------------------------------|---------------------------------------------------------------------|
| $n = 1$  | $\mu = \mathbb{E}[X]$<br>“mean” | 0                                         | 0                                                                   |
| $n = 2$  | $E(X^2)$                        | $\sigma^2 = E[(X - \mu)^2]$<br>“variance” | 1                                                                   |
| $n = 3$  | $E(X^3)$                        | $E[(X - \mu)^3]$                          | $E\left[\left(\frac{X - \mu}{\sigma}\right)^3\right]$<br>“skewness” |
| $n = 4$  | $E(X^4)$                        | $E[(X - \mu)^4]$                          | $E\left[\left(\frac{X - \mu}{\sigma}\right)^4\right]$<br>“kurtosis” |
| $\vdots$ | $\vdots$                        | $\vdots$                                  | $\vdots$                                                            |

Table 1.4

**Definition 1.21.** We say a moment  $E(X^n)$  **exists** if the corresponding absolute moment is finite:  $E(|X|^n) < \infty$ . So the mean exists, for example, if  $E(|X|) < \infty$ .

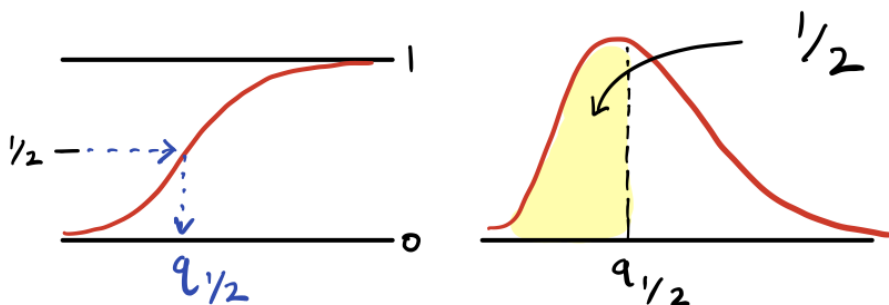


Figure 1.29

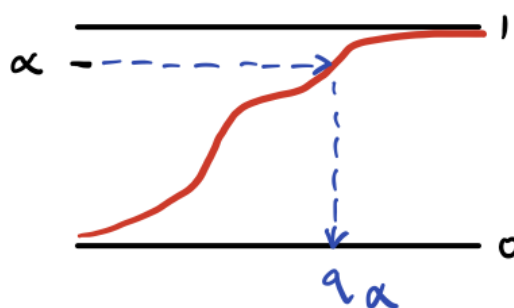


Figure 1.30

## 1.7.2 Quantiles

The mean and standard deviation have the significant drawback that they are not robust to outliers. Consider adding Elon Musk's income to the distribution of incomes in America. This will drastically increase both the mean and the standard deviation! *Quantiles*, on the other hand, are robust to outliers.

**Definition 1.22.** An  $\alpha$ th quantile is any real number  $q_\alpha$  satisfying

$$P(X \leq q_\alpha) = \alpha.$$

In particular, the **median** is the quantile  $q_{1/2}$ .

So  $q_\alpha$  is a point in the range with  $\alpha$  probability to the left of it. Note that quantiles are not unique in general. When the cdf  $F_X$  is an invertible function (no jumps or plateaus), then the quantiles are unique, and you can define them as

$$q_\alpha = F_X^{-1}(\alpha).$$

For this reason, the inverse cdf  $F_X^{-1}$  is called the **quantile function**. See Figure 1.29 and 1.30.

**Example 1.20.** Consider the discrete distribution  $X \sim \text{Unif}(1.2, 3.4, 5, 6.1, 8.9, 10)$ , in which each of these numbers has probability  $1/6$ . Then e.g.  $q_{5/6} = 8.9$ , and the median is  $q_{1/2} = 5$ .

Consider replacing the 10 with 100. This will have a significant effect on the mean. Previously,  $\mathbb{E}[X] = (1.2 + 3.4 + 5 + 6.1 + 8.9 + 10)/6 = 5.767$ . With 100 instead of 10, we have  $\mathbb{E}[X] = (1.2 + 3.4 + 5 + 6.1 + 8.9 + 100)/6 = 20.767$ . But the median remains the same: it's  $q_{1/2} = 5$  either way.

The above example shows that the median is a robust measure of location. What about a robust measure of spread? We can also use quantiles for that.

**Definition 1.23.** The interquartile range (IQR) is defined as

$$\text{IQR} = q_{3/4} - q_{1/4}.$$

In words, the IQR is the width of the interval containing the middle 50% of the data. We summarize these different measures below.

|            | location | spread    |
|------------|----------|-----------|
| not robust | mean     | std. dev. |
| robust     | median   | IQR       |

**Example 1.21.** Let  $X \sim \text{Exp}(\beta)$ . Recall from last lecture that the cdf of  $\text{Exp}(\beta)$  is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\beta x}, & x \geq 0. \end{cases}$$

Let us compute  $q_\alpha$  for some  $\alpha \in (0, 1)$ . We solve

$$\begin{aligned} \alpha &= 1 - e^{-\beta q_\alpha} \\ e^{-\beta q_\alpha} &= 1 - \alpha \\ -\beta q_\alpha &= \ln(1 - \alpha) \\ q_\alpha &= -\frac{1}{\beta} \ln(1 - \alpha) \end{aligned}$$

and so the quantile function is

$$F_X^{-1}(\alpha) = -\frac{1}{\beta} \ln(1 - \alpha).$$

The median of the exponential distribution is

$$F_X^{-1}(0.5) = -\frac{\ln(1 - 0.5)}{\beta} = -\frac{\ln(0.5)}{\beta} = \frac{\ln 2}{\beta}.$$

## 1.8 The moment-generating function

We have referred many times to the fact that objects like the cdf, the pmf, or the pdf fully and uniquely characterize the entire distribution of a random variable. If you have any one of these objects, you *have* the distribution, and you know everything there is to know about it. Moments are an example of an object that does *not* fully and uniquely characterize the distribution of a random variable. Two random variables can have the exact same moments, and yet be wildly different.

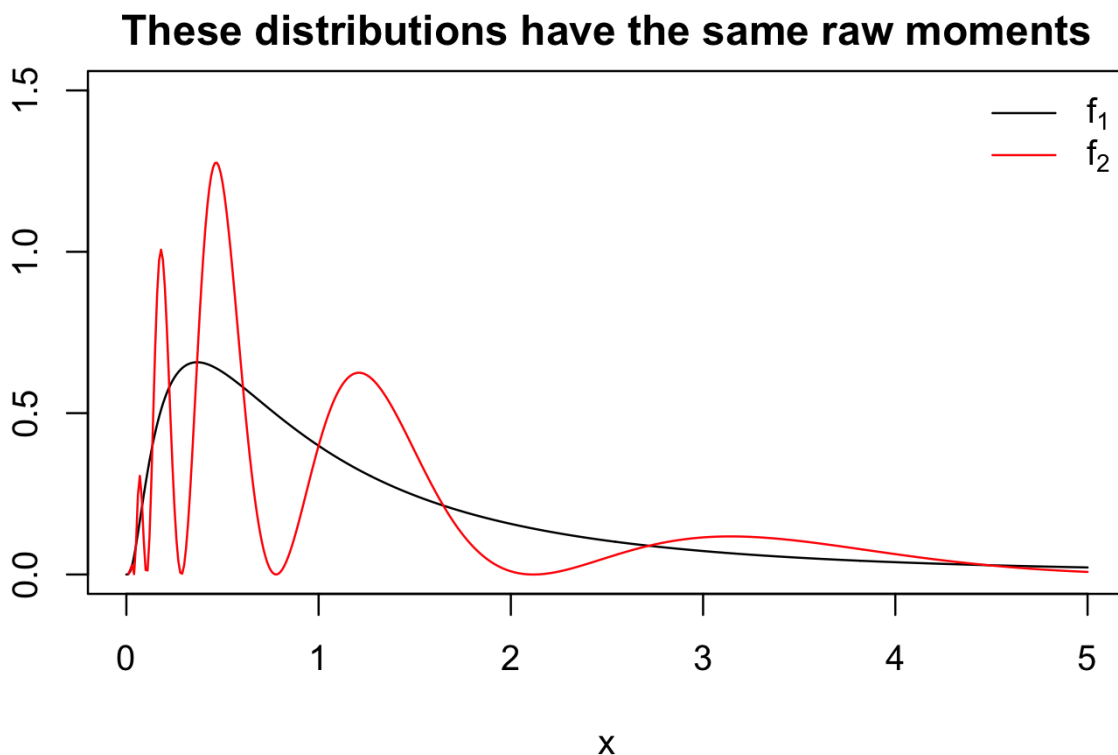


Figure 1.31

**Example 1.22.** Consider  $X \sim N(2, 2)$  and  $Y \sim \text{Poisson}(2)$ . So these distributions are very different (discrete versus continuous, different ranges entirely, etc), and yet  $\mathbb{E}[X] = E(Y) = 2$  and  $\text{var}(X) = \text{var}(Y) = 2$ .

That example shows that the first two moments are not enough to uniquely characterize the distribution, but it turns out that even if *all* of the infinitely-many moments  $\mathbb{E}[X]$ ,  $E(X^2)$ ,  $E(X^3)$ , ... are the same, the distributions could still be very different:

**Example 1.23.** Consider two absolutely continuous random variables  $X_1$  and  $X_2$  with pdfs

$$f_1(x) = \frac{1}{\sqrt{2\pi}} \frac{1}{x} \exp\left(-\frac{1}{2}(\ln x)^2\right), \quad x \geq 0$$

$$f_2(x) = f_1(x)[1 + \sin(2\pi \ln x)], \quad x \geq 0.$$

Figure 1.31 displays these densities, and clearly they are quite different. And yet, you can show

that  $E(X_1^n) = E(X_2^n)$  for all  $n \in \mathbb{N}$ . We start by computing  $E(X_2^n)$  via LOTUS:

$$\begin{aligned}
 E(X_2^n) &= \int_0^\infty x^n f_2(x) dx \\
 &= \int_0^\infty x^n f_1(x) [1 + \sin(2\pi \ln x)] dx \\
 &= \int_0^\infty [x^n f_1(x) + x^n f_1(x) \sin(2\pi \ln x)] dx \\
 &= \int_0^\infty x^n f_1(x) dx + \int_0^\infty x^n f_1(x) \sin(2\pi \ln x) dx \\
 &= E(X_1^n) + \int_0^\infty x^n f_1(x) \sin(2\pi \ln x) dx \\
 &= E(X_1^n) + \int_0^\infty \frac{x^n}{x\sqrt{2\pi}} \exp\left(-\frac{1}{2}(\ln x)^2\right) \sin(2\pi \ln x) dx
 \end{aligned}$$

So it will suffice to show that the second term is zero. We apply the change of variables

$$\begin{aligned}
 u &= \ln x - n \\
 x &= \exp(u + n) \\
 du &= \frac{1}{x} dx.
 \end{aligned}$$

So

$$\begin{aligned}
 E(X_2^n) &= E(X_1^n) + \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} \exp(u + n)^n \exp\left(-\frac{1}{2}(u + n)^2\right) \sin(2\pi(u + n)) du \\
 &= E(X_1^n) + \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} \exp(nu + n^2) \exp\left(-\frac{1}{2}(u + n)^2\right) \sin(2\pi u + 2\pi n) du \\
 &= E(X_1^n) + \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} \exp(nu + n^2) \exp\left(-\frac{1}{2}u^2 - nu - \frac{1}{2}n^2\right) \sin(2\pi u) du \\
 &= E(X_1^n) + \underbrace{\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(u^2 - n^2)\right) \sin(2\pi u) du}_{\text{odd function}} \\
 &= E(X_1^n) + 0 \\
 &= E(X_1^n).
 \end{aligned}$$

Alright alright. So moments, even infinitely many of them, are not enough to uniquely characterize a distribution. But there exists an object intimately related to the moments that does uniquely characterize its distribution:

**Definition 1.24.** If  $X$  is any random variable, and the expected value  $E[e^{tX}]$  exists and is finite for all  $t$  in a neighborhood of zero (so for all  $-b < t < b$  for some  $b > 0$ ), then the **moment-**

**generating function (mgf)** of  $X$  is

$$M_X(t) = E[e^{tX}], \quad t \in (-b, b). \quad (1.30)$$

**Remark 1.14.** The moment generating function might not exist! As we have seen, not all expected values exist or are finite, so not all random variables have a moment generating function.

**Theorem 1.14.** If  $X$  and  $Y$  are random variables with mgfs  $M_X$  and  $M_Y$ , and  $M_X = M_Y$ , then  $X$  and  $Y$  have the exact same distribution.

We will not prove this result, but it means that the mgf, like the cdf or pmf or pdf, is one of those objects that fully and uniquely characterizes the entire distribution of a random variable. If you know the mgf, you know everything.

That's all well and good, but what's the point of this new object? Here's the point:

**Theorem 1.15.** If  $X$  is a random variable with a moment generating function  $M_X$ , then the  $n$ th derivative of the mgf evaluated at zero is equal to the  $n$ th raw moment:

$$M_X^{(n)}(0) = E(X^n) \quad \forall n \in \mathbb{N}. \quad (1.31)$$

*Proof sketch.* We can rewrite  $M_X$  with a Taylor series expansion about zero (ie a Maclaurin series):

$$M_X(t) = \sum_{n=0}^{\infty} \frac{M_X^{(n)}(0)}{n!} t^n.$$

We can also rewrite  $e^{tX}$  using the Taylor series expansion of the exponential function

$$e^{tX} = \sum_{n=0}^{\infty} \frac{(tX)^n}{n!} = \sum_{n=0}^{\infty} \frac{X^n}{n!} t^n.$$

While it is not always the case that the expected value of an infinite sum is the infinite sum of the expected values, it is true in this case (for reasons I will not go into), and so

$$E[e^{tX}] = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!} t^n.$$

By definition, we know that  $M_X(t) = E(e^{tX})$ , and so it must be the case that

$$\sum_{n=0}^{\infty} \frac{M_X^{(n)}(0)}{n!} t^n = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!} t^n.$$

The only way these can be equal if is each term is equal, and so  $M_X^{(n)}(0) = E(X^n)$ . □

**Remark 1.15.** Theorem 1.15 is killer. It allows us to compute moments by taking a derivative instead of simplifying an integral or an infinite series. As a general matter, differentiation is much more straightforward than integration, so this will be very useful. Now that we have this tool, we can use it to do some moment calculations that we've been putting off.



**Remark 1.16.** The moment-generating function is intimately related to something called the *Laplace transform*, which you may encounter in a course on ordinary differential equations.

**Example 1.24.** Let  $X \sim \text{Poisson}(\lambda)$ , and fix any  $t \in \mathbb{R}$ . Then

$$\begin{aligned}
 M_X(t) &= E[e^{tX}] \\
 &= \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} e^{-\lambda} \\
 &= \sum_{k=0}^{\infty} (e^t)^k \frac{\lambda^k}{k!} e^{-\lambda} \\
 &= \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} e^{-\lambda} \\
 &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} \\
 &= e^{-\lambda} e^{\lambda e^t} \\
 &= e^{\lambda e^t - \lambda}.
 \end{aligned}$$

Note that we did not have to place any restrictions on  $t$  to complete the computation, so the mgf is defined for all  $t$  and we have

$$M_X(t) = \exp(\lambda e^t - \lambda), \quad t \in \mathbb{R}. \quad (1.32)$$

We can use this to compute the mean and variance of the Poisson:

$$\begin{aligned}
 M'_X(t) &= \lambda e^t \exp(\lambda e^t - \lambda) \\
 M''_X(t) &= (\lambda e^t)^2 \exp(\lambda e^t - \lambda) + \lambda e^t \exp(\lambda e^t - \lambda) \\
 M'_X(0) &= \mathbb{E}[X] = \lambda \\
 M''_X(0) &= E(X^2) = \lambda^2 + \lambda
 \end{aligned}$$

So  $\mathbb{E}[X] = \lambda$  and  $\text{var}(X) = E(X^2) - \mathbb{E}[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$ . Once you have the mgf, getting these moment calculations done with derivatives is way easier than proceeding from LOTUS and grinding through an infinite series or integral.

**Example 1.25.** Let  $X \sim N(0, 1)$ , and fix any  $t \in \mathbb{R}$ . Then

|                                                                                                                                    |                               |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| $M_X(t) = E[e^{tX}]$                                                                                                               | definition                    |
| $= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$                                                    | LOTUS                         |
| $= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2 + tx} dx$                                                      | combine base- $e$ terms       |
| $= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2 - 2tx)} dx$                                                   | factor out $-1/2$             |
| $= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2 - 2tx + t^2 - t^2)} dx$                                       | add/subtract $t^2$            |
| $= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}[(x-t)^2 - t^2]} dx$                                               | factor first three terms      |
| $= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2 + \frac{1}{2}t^2} dx$                                      | distribute $-1/2$             |
| $= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} e^{\frac{1}{2}t^2} dx$                                    |                               |
| $= e^{\frac{1}{2}t^2} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx}_{N(t, 1) \text{ PDF}}$ | pull constant out of integral |
| $= e^{\frac{1}{2}t^2} \cdot 1$                                                                                                     | PDFs integrate to 1           |
| $= e^{\frac{1}{2}t^2}.$                                                                                                            |                               |

Note two things:

- We did not have to place any restrictions on  $t$  to complete the computation, so the mgf is defined for all  $t$ ;
- This is a classic example of “massage and squint.” We just pushed and pulled the integrand and massaged it with constants until it turned into something familiar. Then we applied an identity.

So the mgf of the standard normal is

$$M_X(t) = \exp\left(\frac{1}{2}t^2\right), \quad t \in \mathbb{R}. \quad (1.33)$$

We can use this to compute the mean and variance of the standard normal:

$$M'_X(t) = t \exp\left(\frac{1}{2}t^2\right)$$

$$M''_X(t) = t^2 \exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}t^2\right)$$

$$M'_X(0) = \mathbb{E}[X] = 0$$

$$M''_X(0) = E(X^2) = 0 + 1 = 1$$

So  $\mathbb{E}[X] = 0$  and  $\text{var}(X) = E(X^2) - \mathbb{E}[X]^2 = 1 - 0 = 1$ . Not exactly earth-shattering, but note that it's the first time we've actually seen a calculation of these facts.

**Theorem 1.16.** If the moment generating function of a random variable  $X$  exists, then all of its moments are finite:  $E[|X|^n] < \infty$  for all  $n \in \mathbb{N}$ .

However, the converse is not true. A random variable could have all finite moments, and yet the mgf does not exist. So all moments finite is a *necessary condition* for the existence of the moment generating function, but not a sufficient one.

### Summary of basic facts about moments and mgfs

- mgf exists **does imply**  $E(X^n)$  finite for all  $n \in \mathbb{N}$ ;
- If mgf exists, then  $M_X^{(n)}(0) = E(X^n)$  for all  $n \in \mathbb{N}$ ;
- $E(X^n)$  finite for all  $n \in \mathbb{N}$  **does not imply** mgf exists;
- $M_X = M_Y$  **does imply**  $F_X = F_Y$ ;
- $E(X^n) = E(Y^n)$  for all  $n \in \mathbb{N}$  **does not imply**  $F_X = F_Y$ .